CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

May 2012

CL209 Structural Analysis-I

Date: 16.05.2012, Wednesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Differentiate between Statically Determinate & Indeterminate structure. [04]

(b) What do you understand by stability of structure? [03]

Q - 2 (a) Prove in case of a thin shell, Circumferential stress, $f_c = Pd/2t$ and Longitudinal stress, $f_l = Pd/4t$ [04]

(b) **Attempt any two:** [10]

(i) A short hollow pier of 1.2 m square section outside and 1.0 m square inside is subjected to a direct load of 120 kN along its outer edge point. Determine the final stress at the base of pier. Draw neat sketch of stress distribution diagram.

(ii) Find the maximum and minimum stress intensities in a short C.I. Column of hollow circular section attached with a projection of bracket carrying a load of 60 kN. The external and internal diameters of the column are 300 mm and 250 mm respectively. The load line is off the axis of the column by the 300 mm.

(iii) Find the maximum and minimum stress intensities induced on the base of a masonary wall 6 meter high, 4 m wide and 1.5 m thick subjected to a horizontal wind pressure of 1500 N/mm^2 acting on 4 meter side. The density of masonary may be taken as 22400 N/mm^3.

Q - 3 (a) A watermain 500 mm diameter contains water at a pressure head of 100 m. If the weight of water is 10 kN/m^3 and allowable stress in the pipe 20 N/mm^2, calculate the thickness of the pipe. [04]

(b) **Attempt any two:** [10]

(i) A rectangle column section, size B x D is fixed at both ends. Determine the limiting length of the column so that critical stress is 7 N/mm^2. Take E = 20000 N/mm^2.

(ii) A slender strut, 2100 mm long and of rectangular section 45mm x 15mm transmits a longitudinal load P acting at the centre of each end. The strut was slightly bent about its minor principal axis before loading. If P is increased from 750N to 2000N, the deflection at the middle of the length increases by 5 mm. Determine the amount of deflection before loading. Find also the total deflection and the maximum stress when P is 2500N.Take $E = 2 \times 10^5$ N/mm^2.

(iii) A cast iron column of hollow cylindrical section 6.5 meters long, with ends firmly built-in, has to carry an axial load of 375 kN. Determine the section, using a factor of safety of 6. Internal diameter to be 0.55 of the external diameter. Rankine's constants for C.I. are $f_c = 550$ N/mm^2 and a = 1/1600.

SECTION – II

Q - 4 (a) A three hinged circular arch of span 21 m has a rise of 4 m. The arch is loaded with a point load 8 kN at a horizontal distance of 6 m from the left support. Determine the horizontal thrust, two reactions and bending moment under the load. [04]

(b) A three hinged arch of span 40 m and rise 8 m carries concentrated loads of 200 kN and 150 kN at distance of 8 m and 16 m from the left end and an UDL of 50 kN/m on the right half of the span. Find the horizontal thrust. [03]

Q - 5 (a) A three hinged arch of span 'l' and rise 'h' carries a uniformly distributed load of 'w' per unit run over the whole span. Show that the horizontal thrust at each support is $wl^2/8h$. [04]

(b) **Attempt any two:** [10]

(i) Derive the differential equation of an elastic curve.

(ii) Using double integration method, obtain slopes at supports and deflection at the centre of a simply supported beam of length 4 m subjected to a central point load of 240 kN. Take c/s of beam = 100 mm x 200 mm and E = 0.7 x 10^5 N/mm^2.

(iii) A simply supported beam AB of length of 6 m is loaded with three point loads of 30 kN, 50 kN and 70 kN at distance 1 m, 3 m and 5 m from end A. Determine the position and magnitude of maximum deflection. Take E = 2*10^5 N/mm^2 and I = 9200 cm^4.

Q - 6 (a) List out methods for determining slope and deflection at any point of a beam. [02]

(b) **Attempt any two:** [12]

(i) Two wheel loads of 16kN and 8kN at a fixed distance 2m apart cross a beam of 10m span. Draw the influence line diagrams for B.M. and S.F. for a point 4m from the left abutment. Also find maximum B.M and S.F. at that point.

(ii) Four wheel loads of 6, 4, 8 and 5 kN crosses a girder of 20m span from left to right followed by u.d.l. of 4kN/m and 4m long with the 6kN load leading. The spacing between the loads in same order are 3m, 2m and 2m. The head of the u.d.l. is at lm from the last 5 kN load. Using influence lines, Calculate S.F. and B.M. at a section 8m from the left support when the 4kN load is at centre of the span.

(iii) Two wheel loads, 50 kN each spaced at 4m centre to centre, cross a girder of 8m span. Find absolute maximum B.M. in the beam.

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